

Software Requirements Specification

Renewal & Upsell Advisor

Agent ID: S-007

Suite Category: Sales Suite

Version: 1.0

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Confidential

Contents

1	Project Overview	5
1.1	Project Title	5
1.2	Agent Identification	5
1.3	Executive Summary	5
2	Objectives	5
2.1	Primary Objectives	5
2.2	Secondary Objectives	6
3	Scope	6
3.1	In-Scope	6
3.2	Out-of-Scope	7
4	Business Scenario	7
4.1	Use Case Context	7
4.2	Primary Business Scenario	7
5	Milestones & Timeline	9
5.1	7-Day Development Sprint	9
5.2	Detailed Task Breakdown	9
5.3	Critical Path Dependencies	11
6	Roles & Responsibilities	11
6.1	Technical Team Structure	11
6.2	Team Size	12
7	Deliverables	12
7.1	Technical Deliverables	12
7.2	Documentation Deliverables	14
8	Tech Stack	14
8.1	Frontend Technologies	14
8.2	Backend Technologies	15
8.3	Machine Learning Stack	15
8.4	Database & Storage	15
8.5	Integration & APIs	16
8.6	DevOps & Infrastructure	16
8.7	Development Tools	16
9	Functional Requirements	17
9.1	FR-1: Contract & Renewal Tracking	17
9.1.1	FR-1.1: Contract Data Ingestion	17
9.1.2	FR-1.2: Renewal Timeline Monitoring	17
9.2	FR-2: Churn Risk Prediction	17
9.2.1	FR-2.1: Risk Scoring Model	17
9.2.2	FR-2.2: Risk Segmentation	17
9.2.3	FR-2.3: Early Warning Alerts	18

9.3	FR-3: Upsell & Cross-sell Identification	18
9.3.1	FR-3.1: Expansion Opportunity Detection	18
9.3.2	FR-3.2: Cross-sell Recommendations	18
9.3.3	FR-3.3: Opportunity Qualification	18
9.4	FR-4: Automated Playbook Recommendations	18
9.4.1	FR-4.1: Playbook Library	18
9.4.2	FR-4.2: Playbook Matching Engine	19
9.4.3	FR-4.3: Personalized Messaging	19
9.5	FR-5: Outreach Cadence Optimization	19
9.5.1	FR-5.1: Multi-Channel Sequencing	19
9.5.2	FR-5.2: Task Automation	19
9.6	FR-6: Sentiment & Engagement Analysis	20
9.6.1	FR-6.1: Email Sentiment Tracking	20
9.6.2	FR-6.2: Engagement Monitoring	20
9.7	FR-7: Analytics & Reporting	20
9.7.1	FR-7.1: Renewal Pipeline Dashboard	20
9.7.2	FR-7.2: Expansion Revenue Tracking	20
9.7.3	FR-7.3: Agent Performance Metrics	20
9.7.4	FR-7.4: Executive Reports	21
9.8	FR-8: Integration Requirements	21
9.8.1	FR-8.1: Salesforce Integration	21
9.8.2	FR-8.2: Billing Platform Integration	21
9.8.3	FR-8.3: Analytics Platform Integration	21
9.9	FR-9: Notification System	21
9.9.1	FR-9.1: Alert Delivery	21
9.9.2	FR-9.2: Alert Types	21
9.10	FR-10: Security & Compliance	22
9.10.1	FR-10.1: Authentication & Authorization	22
9.10.2	FR-10.2: Data Privacy	22
9.11	FR-11: Autonomous Negotiation & Execution	22
9.11.1	FR-11.1: Autonomous Engagement	22
9.11.2	FR-11.2: Negotiation Logic	22
9.11.3	FR-11.3: Contract Execution	22
10	Flow Diagram / Architecture Diagram	23
10.1	System Architecture Overview	23
10.2	Data Flow	23
10.3	Architecture Components	23
10.4	Detailed Flow Descriptions	24
10.4.1	Flow 1: Churn Risk Detection	24
10.4.2	Flow 2: Upsell Opportunity Identification	25
10.4.3	Flow 3: Renewal Forecast Generation	25
11	Competitive Analysis	25
11.1	Market Landscape	25
11.2	Primary Competitors	26
11.2.1	Gainsight Customer Success	26
11.2.2	ChurnZero	26
11.2.3	Yagna iQ Channel Ecosystem Platform	27

11.2.4	Totango	27
11.2.5	Catalyst (formerly ProsperWorks)	28
11.3	Innovation & Differentiation Strategy	28
11.3.1	Core Innovation: The "Agentic" Shift	28
11.3.2	Our Competitive Advantages	29
11.4	Feature Comparison Matrix	29
11.5	Market Presence & Feedback	29
11.6	References	30
12	Assumptions & Constraints	30
12.1	Assumptions	30
12.2	Constraints	31
13	Success Metrics	31
13.1	Technical KPIs	31
13.2	Business KPIs	31
14	Risk Management	31
14.1	Technical Risks	31
14.2	Business Risks	32
15	Appendices	32
15.1	Appendix A: Glossary	32
15.2	Appendix B: API Endpoints (Sample)	32
15.3	Appendix C: Contact Information	32
16	Document Approval	33

1 Project Overview

1.1 Project Title

Renewal & Upsell Advisor - AI-Powered Revenue Protection and Expansion Agent

1.2 Agent Identification

- **Agent ID:** S-007
- **Agent Name:** Renewal & Upsell Advisor
- **Suite Category:** Sales Suite
- **Version:** 1.0

1.3 Executive Summary

The Renewal & Upsell Advisor is an intelligent AI agent designed to monitor customer contracts, usage patterns, and sentiment signals to predict renewal risk and identify upsell opportunities. The agent provides actionable recommendations including renewal plays, outreach cadences, and personalized value messaging to protect Annual Recurring Revenue (ARR) and maximize expansion revenue. By leveraging machine learning algorithms and real-time data integration from CRM, billing, and analytics platforms, the agent enables sales and customer success teams to take proactive measures before churn occurs.

2 Objectives

2.1 Primary Objectives

1. **Renewal Risk Prediction:** Implement predictive models to identify accounts at risk of churn 60-90 days before contract expiration with 85% accuracy.
2. **Upsell Opportunity Detection:** Automatically surface high-probability upsell and cross-sell opportunities based on product usage, contract value, and customer behavior patterns.
3. **Automated Playbook Execution:** Generate and execute personalized renewal and expansion playbooks with recommended outreach timing, messaging, and engagement strategies.
4. **ARR Protection:** Reduce revenue churn by 20-30% through early intervention and proactive customer engagement.
5. **Revenue Expansion:** Increase expansion revenue by 15-25% by identifying and converting qualified upsell opportunities.

2.2 Secondary Objectives

1. Enable real-time visibility into renewal pipeline health and revenue forecasting.
2. Reduce manual effort for sales and CS teams by automating renewal tracking and risk monitoring.
3. Provide data-driven insights to optimize pricing and packaging strategies.
4. Integrate sentiment analysis from customer communications to enhance prediction accuracy.
5. Generate automated renewal reports and executive dashboards.

3 Scope

3.1 In-Scope

- **Renewal Risk Management:**

- Contract expiration tracking and automated alerts
- Multi-factor churn prediction using ML models
- Customer health scoring based on usage, engagement, and sentiment
- Risk segmentation (High, Medium, Low) with confidence scores

- **Upsell & Cross-sell Intelligence:**

- Product adoption and feature usage analysis
- Revenue expansion opportunity identification
- Cross-sell recommendations based on customer profile and behavior
- Whitespace analysis for untapped product offerings

- **Automated Playbook Recommendations:**

- Personalized renewal and expansion plays
- Multi-channel outreach cadence suggestions (Email, Phone, Meeting)
- Value messaging templates tailored to customer pain points and outcomes
- Best-action recommendations with success probability scores

- **Integration Capabilities:**

- Primary: Salesforce CRM, Stripe/Chargebee billing platforms, Google Analytics/Mixpanel
- Secondary: Customer Success platforms (Gainsight, ChurnZero), Email (Gmail, Outlook), Product analytics tools
- Webhook-based real-time data synchronization

- **Analytics & Reporting:**

- Renewal forecast dashboard with ARR projections
- Pipeline health metrics and trend analysis
- Agent performance tracking and recommendation outcomes
- Executive summary reports with KPI visualization

3.2 Out-of-Scope

- Contract negotiation and legal document generation
- Direct payment processing and transaction handling
- Customer onboarding and implementation workflows
- Product roadmap planning and feature prioritization
- Marketing campaign creation and execution
- Technical support ticket management
- Full financial accounting and revenue recognition
- Integration with non-B2B SaaS business models (e.g., e-commerce, retail)
- Manual override of contract terms or pricing
- Real-time voice call monitoring and transcription

4 Business Scenario

4.1 Use Case Context

Enterprise SaaS companies typically face 5-7% annual revenue churn and miss 30-40% of expansion opportunities due to reactive renewal processes. Traditional methods rely on human CSMs to identify risks and negotiate renewals manually, which is unscalable for long-tail accounts. Sales teams often lack visibility into real-time customer health signals, leading to missed upsell windows.

4.2 Primary Business Scenario

Scenario: Fully Autonomous Renewal Negotiation and Upsell Execution

Actors:

- **Autonomous Revenue Agent (S-007)**
- Customer Decision Maker (Signer)
- Revenue Operations Manager (Supervisor)
- Account Executive (AE) - Override Only

Example Workflow:

Day 0-90 Before Renewal - Autonomous Monitoring:

1. TechCorp has 150 enterprise customers renewing in 90 days (\$8.5M ARR).
2. The Agent continuously monitors real-time signals: contract terms (Salesforce), payment health (Stripe), usage trends (Mixpanel), and support sentiment (Zendesk).
3. The Agent identifies 12 High-Risk accounts. For Account XYZ (\$180K ARR), it detects low login frequency (-45%) and negative sentiment in recent tickets.

Day 1-30 - Autonomous Negotiation (Risk Mitigation):

1. **Agent Action:** Instead of alerting a CSM to manually email, the Agent autonomously drafts and sends a personalized "Value Realization" email to the XYZ champion.
2. **Agent Capability:** The email highlights specific ROI metrics achieved ("Your team saved 400 hours using Feature A despite lower logins") and proposes a check-in call.
3. **Negotiation:** The customer replies with pricing concerns. The Agent, adhering to pre-approved discount guardrails, autonomously negotiates: "We can offer a 5% loyalty discount for a 2-year early renewal."
4. **Outcome:** The customer accepts via email. The Agent updates the Salesforce Opportunity to "Verbal Agreement" and triggers the contract generation workflow.

Day 30-60 - Autonomous Upsell Pitching:

1. **Opportunity Detection:** For healthy Account ABC (\$250K ARR), the Agent detects 92% license utilization and high usage of the "Standard Reporting" feature.
2. **Agent Action:** The Agent identifies a "Premium Analytics" upsell opportunity (\$75K expansion). It autonomously sends a "Growth Proposal" to the ABC admin.
3. **Pitch:** "You are hitting limits on Standard Reports. Upgrading to Premium will unlock unlimited history and predictive charts. Click here to activate a 14-day trial or instant upgrade."
4. **Closing:** The admin clicks "Instant Upgrade". The Agent processes the prorated invoice via Stripe and provisions the licenses automatically.

Day 60-90 - Execution and Closure:

1. The Agent finalizes all renewal contracts via DocuSign integration.
2. For complex deals requiring human sign-off (e.g., >\$500K or non-standard terms), the Agent loops in the VP of Sales with a summarized "Deal Memo" for one-click approval.
3. Revenue Operations receives a finalized forecast: "Closed Renewals: \$8.1M (95% retention), Expansion Booked: \$920K".

Business Outcome:

- **Reduced Churn:** 40% reduction in churn via instant, 24/7 autonomous negotiation.

- **Increased Efficiency:** 70% reduction in CSM manual overhead; CSMs focus only on strategic relationships.
- **Expansion Revenue:** 25% increase in upsell conversion due to timely, usage-based autonomous offers.
- **Faster Cycles:** Renewal cycle shortened from 45 days to 14 days.

5 Milestones & Timeline

5.1 7-Day Development Sprint

Overall Duration: 7 days (168 hours)

Sprint Goal: Deploy production-ready Renewal & Upsell Advisor agent with core ML models, integration pipelines, and automated playbook engine.

5.2 Detailed Task Breakdown

Day	Tasks	Duration
Day 1	Architecture & Setup - Environment provisioning (AWS/GCP infrastructure) - Database schema design (PostgreSQL/MongoDB) - API gateway configuration - Authentication & authorization setup - CI/CD pipeline establishment - Development environment configuration	8 hours
Day 1-2	Integration Layer Development - Salesforce API integration (REST/SOAP) - Stripe/Chargebee webhook setup - OAuth 2.0 authentication flows - Data extraction pipelines (contracts, billing, accounts) - Real-time event listeners - Error handling & retry mechanisms - Analytics platform connectors (Segment, Mixpanel)	14 hours
Day 2-3	Data Pipeline & Feature Engineering - ETL pipeline development - Data normalization & cleansing - Feature extraction engine: - Contract metadata (renewal date, ARR, tenure) - Usage metrics (login frequency, feature adoption) - Behavioral signals (support tickets, NPS) - Payment history (late payments, failed charges) - Time-series aggregation (7/30/90-day windows) - Feature store implementation	12 hours

Day 3-4	ML Model Development - Churn prediction model (Gradient Boosting/Random Forest) - Training on historical renewal outcomes - Model evaluation (Precision, Recall, F1-score) - Hyperparameter tuning - Upsell propensity scoring model - Cross-sell recommendation engine - Model versioning & registry setup - Inference API endpoints	16 hours
Day 4-5	Playbook Engine & Recommendation System - Rule-based playbook matching logic - Outreach cadence optimizer - Value messaging template engine - Personalization layer (dynamic variables) - Recommendation scoring algorithm - Action prioritization framework - A/B testing infrastructure for playbooks	14 hours
Day 5-6	Notification & Workflow Automation - Slack/Teams webhook integration - Email notification service (SendGrid/AWS SES) - In-app alert system - Task automation in Salesforce (create opportunities, tasks) - Scheduled batch jobs (daily risk scoring) - Event-driven trigger system - Workflow orchestration (Apache Airflow)	12 hours
Day 6	Dashboard & Analytics UI - Renewal pipeline dashboard (React/Vue.js) - Risk heatmap visualization - ARR forecasting charts - Agent performance metrics - Drill-down account details view - Export functionality (CSV, PDF) - Responsive design implementation	10 hours
Day 7	Testing, Deployment & Documentation - Unit testing (Jest, PyTest) - Integration testing (API contract validation) - End-to-end scenario testing - Performance testing (load, stress) - Security testing (penetration, vulnerability scan) - Production deployment - Rollback procedures - Technical documentation - User guide & playbook library	14 hours
Total Development Hours:		100 hours

5.3 Critical Path Dependencies

1. Integration Layer must complete before Data Pipeline development
2. Feature Engineering required before ML Model training
3. ML Models must be operational before Playbook Engine development
4. All backend services completed before Dashboard UI build

6 Roles & Responsibilities

6.1 Technical Team Structure

Role	Responsibilities
Tech Lead / Architect	- System architecture design & technical decision-making - Technology stack selection - Integration strategy definition - Code review & quality assurance - Team coordination & sprint planning - Risk identification & mitigation - Performance optimization & scalability planning
Backend Engineer 1 (Integrations)	- Salesforce API integration development - Stripe/Chargebee webhook implementation - OAuth authentication flows - REST API endpoint development - Data synchronization pipelines - Error handling & logging - API documentation (Swagger/OpenAPI)
Backend Engineer 2 (Data Pipeline)	- ETL pipeline architecture & implementation - Data transformation & normalization logic - Feature engineering workflows - Database schema design & optimization - Batch processing jobs (Apache Airflow) - Data quality validation & monitoring - Time-series data aggregation
ML Engineer	- ML model development (churn prediction, upsell scoring) - Feature selection & dimensionality reduction - Model training, validation & hyperparameter tuning - ML pipeline orchestration (MLflow, Kubeflow) - Model deployment & serving infrastructure - A/B testing framework for model versions - Model performance monitoring & retraining triggers

Full-Stack Developer	- Frontend dashboard development (React/Vue.js) - Backend API integration for UI - Data visualization components (Chart.js, D3.js) - Responsive design implementation - User authentication & session management - Playbook recommendation UI - Export functionality (reports, CSV)
DevOps Engineer	- Cloud infrastructure setup (AWS/GCP/Azure) - Kubernetes cluster provisioning & management - CI/CD pipeline configuration (Jenkins, GitHub Actions) - Container orchestration (Docker, Kubernetes) - Monitoring & alerting setup (Prometheus, Grafana) - Secrets management (Vault, AWS Secrets Manager) - Production deployment & rollback procedures - Infrastructure-as-Code (Terraform, CloudFormation)
QA Engineer	- Test plan creation & execution - Unit test development (Jest, PyTest) - Integration testing (API contracts, data flows) - End-to-end scenario testing (Selenium, Cypress) - Performance & load testing (JMeter, Locust) - Security testing coordination - Bug tracking & regression testing - Test automation framework setup
Data Scientist (Consultant)	- Churn prediction model design & algorithm selection - Statistical analysis of renewal patterns - Feature importance analysis - Model interpretability (SHAP, LIME) - Business metrics validation - Recommendation logic optimization - Training data labeling strategy

6.2 Team Size

- **Core Team:** 7 members (Tech Lead, 2 Backend Engineers, 1 ML Engineer, 1 Full-Stack Developer, 1 DevOps Engineer, 1 QA Engineer)
- **Extended Support:** 1 Data Scientist (part-time consultant)

7 Deliverables

7.1 Technical Deliverables

1. Renewal & Upsell Advisor Agent - Production Application

- Dockerized microservices architecture
- Deployed on cloud infrastructure (AWS/GCP)
- Auto-scaling configuration

- 99.5% uptime SLA

2. Machine Learning Models

- Churn prediction model (Gradient Boosting Classifier)
- Upsell propensity scoring model
- Cross-sell recommendation engine
- Model artifacts & versioning (MLflow registry)
- Inference API endpoints

3. Integration Connectors

- Salesforce bidirectional sync
- Stripe/Chargebee webhook listeners
- Google Analytics / Mixpanel event streaming
- Customer Success platform connectors (optional: Gainsight, ChurnZero)
- Email service integration (Gmail, Outlook via MS Graph API)

4. Automated Playbook Engine

- 15+ pre-built renewal & upsell playbooks
- Personalized messaging template library
- Outreach cadence optimizer
- Multi-channel recommendation system

5. Analytics Dashboard

- Renewal pipeline overview
- Risk heatmap & account segmentation
- ARR forecast visualization
- Upsell opportunity tracker
- Agent performance metrics
- Exportable reports (PDF, CSV)

6. Notification System

- Slack/Teams integration
- Email alerts (SendGrid/AWS SES)
- In-app notification center
- Configurable alert rules

7. API Documentation

- REST API specification (OpenAPI 3.0)
- Authentication guide
- Webhook payload schemas
- Code examples (Python, JavaScript)

7.2 Documentation Deliverables

1. Technical Documentation

- System architecture diagram
- Database schema documentation
- API reference guide
- Deployment runbook
- Disaster recovery procedures

2. User Documentation

- User manual (Sales & CS teams)
- Playbook customization guide
- Dashboard navigation tutorial
- Integration setup guide

3. Training Materials

- Video walkthrough (15 minutes)
- Quick start guide
- FAQ document
- Best practices for renewal management

8 Tech Stack

8.1 Frontend Technologies

- **Framework:** React 18.x or Vue.js 3.x
- **State Management:** Redux Toolkit / Vuex
- **UI Library:** Material-UI / Ant Design
- **Data Visualization:** Chart.js, Recharts, D3.js
- **HTTP Client:** Axios
- **Build Tool:** Vite / Webpack 5
- **Testing:** Jest, React Testing Library, Cypress

8.2 Backend Technologies

- **Primary Language:** Python 3.11+
- **Web Framework:** FastAPI / Django REST Framework
- **Task Queue:** Celery with Redis broker
- **Workflow Orchestration:** Apache Airflow
- **Async Processing:** asyncio, aiohttp
- **API Standards:** REST, GraphQL (optional)
- **Authentication:** OAuth 2.0, JWT
- **API Documentation:** Swagger/OpenAPI 3.0

8.3 Machine Learning Stack

- **ML Framework:** Scikit-learn, XGBoost, LightGBM
- **Deep Learning:** TensorFlow / PyTorch (optional for advanced NLP)
- **LLM Service:** OpenAI GPT-4 / Anthropic Claude / AWS Bedrock (for generation)
- **Agent Framework:** LangChain / AutoGen (for orchestration)
- **NLP:** spaCy, Hugging Face Transformers (sentiment analysis)
- **Feature Store:** Feast or custom implementation
- **Model Management:** MLflow
- **Model Serving:** TensorFlow Serving / TorchServe
- **Experimentation:** Weights & Biases (W&B)

8.4 Database & Storage

- **Primary Database:** PostgreSQL 15+ (relational data)
- **NoSQL Database:** MongoDB / DynamoDB (event logs, unstructured data)
- **Caching Layer:** Redis 7+ (session, API response cache)
- **Data Warehouse:** Snowflake / Google BigQuery (analytics)
- **Object Storage:** AWS S3 / Google Cloud Storage (model artifacts, reports)
- **Vector Database:** Pinecone / Weaviate (for similarity search, optional)

8.5 Integration & APIs

- **Salesforce:** REST API, SOAP API, Bulk API, Streaming API
- **Stripe:** REST API, Webhooks
- **Chargebee:** REST API, Webhooks
- **Google Analytics:** Measurement Protocol, Reporting API
- **Mixpanel:** Export API, Engage API
- **Slack:** Web API, Webhook
- **Microsoft Teams:** Graph API, Incoming Webhooks
- **Email:** SendGrid, AWS SES, Gmail API, MS Graph API
- **Customer Success Platforms:** Gainsight API, ChurnZero API

8.6 DevOps & Infrastructure

- **Cloud Platform:** AWS / Google Cloud Platform / Azure
- **Containerization:** Docker
- **Orchestration:** Kubernetes (EKS, GKE, AKS)
- **CI/CD:** GitHub Actions / Jenkins / GitLab CI
- **Infrastructure-as-Code:** Terraform / AWS CloudFormation
- **Monitoring:** Prometheus, Grafana, Datadog
- **Logging:** ELK Stack (Elasticsearch, Logstash, Kibana) / Splunk
- **Error Tracking:** Sentry
- **Secrets Management:** AWS Secrets Manager / HashiCorp Vault
- **Load Balancer:** AWS ALB / NGINX

8.7 Development Tools

- **Version Control:** Git, GitHub/GitLab
- **IDE:** VS Code, PyCharm, Jupyter Notebook
- **API Testing:** Postman, Insomnia
- **Collaboration:** Jira, Confluence, Slack
- **Code Quality:** SonarQube, ESLint, Pylint, Black (formatter)

9 Functional Requirements

9.1 FR-1: Contract & Renewal Tracking

9.1.1 FR-1.1: Contract Data Ingestion

- System SHALL ingest contract data from Salesforce including: contract ID, account name, ARR/MRR, start date, end date, renewal date, contract terms
- System SHALL automatically sync contract updates within 5 minutes of changes in source system
- System SHALL support bulk import of historical contract data (minimum 5 years)

9.1.2 FR-1.2: Renewal Timeline Monitoring

- System SHALL track all contracts with renewal dates within 90-day window
- System SHALL automatically create renewal opportunity records in Salesforce 90 days before expiration
- System SHALL categorize renewals by time horizon: Immediate (0-30 days), Near-term (31-60 days), Future (61-90 days)

9.2 FR-2: Churn Risk Prediction

9.2.1 FR-2.1: Risk Scoring Model

- System SHALL calculate renewal risk score (0-100) for each account using ML model
- Risk score SHALL incorporate features: product usage (login frequency, feature adoption), support metrics (ticket volume, resolution time), payment history (late payments, failed charges), engagement (email opens, meeting attendance), sentiment (NPS, survey responses)
- System SHALL update risk scores daily via batch processing
- System SHALL provide 85% prediction accuracy (validated on historical data)

9.2.2 FR-2.2: Risk Segmentation

- System SHALL classify accounts into risk categories:
 - High Risk: Score 70-100 (churn probability >50%)
 - Medium Risk: Score 40-69 (churn probability 25-50%)
 - Low Risk: Score 0-39 (churn probability <25%)
- System SHALL display top 5 risk factors for each account
- System SHALL provide confidence intervals for predictions

9.2.3 FR-2.3: Early Warning Alerts

- System SHALL send real-time alerts when account moves to High Risk category
- Alerts SHALL include: account name, ARR at risk, risk score, primary risk factors, recommended actions
- System SHALL support alert routing rules (assign to CSM, AE, or team lead based on account ownership)

9.3 FR-3: Upsell & Cross-sell Identification

9.3.1 FR-3.1: Expansion Opportunity Detection

- System SHALL identify upsell opportunities based on:
 - User license utilization (threshold: >80% of purchased seats)
 - Feature usage patterns (high engagement with premium features on basic plan)
 - API usage limits (approaching rate limits)
 - Storage consumption (exceeding 75% of allocated quota)
- System SHALL calculate expansion revenue potential for each opportunity
- System SHALL rank opportunities by expected value and win probability

9.3.2 FR-3.2: Cross-sell Recommendations

- System SHALL recommend complementary products based on:
 - Current product usage (collaborative filtering)
 - Industry segment and company size
 - Similar customer adoption patterns
- System SHALL support whitespace analysis (identify unused products in customer's plan tier)

9.3.3 FR-3.3: Opportunity Qualification

- System SHALL assign propensity score (0-100) to each upsell opportunity
- System SHALL filter opportunities with score >60 for proactive outreach
- System SHALL estimate deal size and expected close date

9.4 FR-4: Automated Playbook Recommendations

9.4.1 FR-4.1: Playbook Library

- System SHALL include minimum 15 pre-built playbooks:
 - Renewal Playbooks: High-Risk Intervention, Executive Business Review, Value Realization Workshop, Competitive Displacement Defense

- Upsell Playbooks: License Expansion, Feature Upgrade, Multi-Year Commitment, Premium Support Add-on
- Each playbook SHALL contain: trigger conditions, outreach cadence (timeline), email/call script templates, success metrics

9.4.2 FR-4.2: Playbook Matching Engine

- System SHALL automatically recommend playbook based on: account risk level, contract value, industry vertical, customer lifecycle stage
- System SHALL support A/B testing of playbook effectiveness
- System SHALL allow CSM/AE to override recommendations

9.4.3 FR-4.3: Personalized Messaging

- System SHALL generate personalized email templates with dynamic variables:
 - Customer-specific ROI metrics
 - Recent usage statistics
 - Relevant case studies from similar companies
 - Competitive insights (if applicable)
- Templates SHALL maintain brand voice and compliance requirements

9.5 FR-5: Outreach Cadence Optimization

9.5.1 FR-5.1: Multi-Channel Sequencing

- System SHALL recommend outreach sequence across channels: Email, Phone Call, LinkedIn Message, In-Product Notification
- System SHALL optimize timing based on: historical engagement patterns, recipient time zone, industry benchmarks
- System SHALL support customizable cadence templates (3-touch, 5-touch, 7-touch sequences)

9.5.2 FR-5.2: Task Automation

- System SHALL automatically create tasks in Salesforce for recommended actions
- System SHALL set due dates aligned with playbook timeline
- System SHALL send reminders 24 hours before task due date

9.6 FR-6: Sentiment & Engagement Analysis

9.6.1 FR-6.1: Email Sentiment Tracking

- System SHALL analyze sentiment from customer emails using NLP (spaCy, Hugging Face)
- System SHALL classify sentiment as Positive, Neutral, Negative with confidence score
- System SHALL detect sentiment shifts (declining trend over 30-day window)

9.6.2 FR-6.2: Engagement Monitoring

- System SHALL track engagement metrics: email open rate, link click rate, meeting acceptance rate, response time
- System SHALL flag disengaged accounts (no response to 3+ outreach attempts)

9.7 FR-7: Analytics & Reporting

9.7.1 FR-7.1: Renewal Pipeline Dashboard

- Dashboard SHALL display:
 - Total ARR at risk (segmented by risk level)
 - Renewal rate forecast (current quarter, next quarter)
 - Top 10 at-risk accounts by ARR
 - Churn reason analysis (historical)
- Dashboard SHALL support filtering by: renewal date range, account segment, assigned CSM/AE

9.7.2 FR-7.2: Expansion Revenue Tracking

- Dashboard SHALL display:
 - Total expansion pipeline value
 - Upsell conversion rate
 - Average deal size by opportunity type
 - Time-to-close for expansion deals

9.7.3 FR-7.3: Agent Performance Metrics

- System SHALL track:
 - Prediction accuracy (renewal outcome vs. predicted risk)
 - Playbook effectiveness (conversion rate by playbook type)
 - Alert response time (time from alert to first action)
 - Revenue saved via early intervention

9.7.4 FR-7.4: Executive Reports

- System SHALL generate weekly executive summary including: ARR retention forecast, Top risks & mitigations, Expansion opportunities closed, YoY performance trends
- Reports SHALL be exportable in PDF and CSV formats

9.8 FR-8: Integration Requirements

9.8.1 FR-8.1: Salesforce Integration

- System SHALL sync bidirectionally: Accounts, Opportunities, Contracts, Tasks, Activities
- System SHALL support custom Salesforce objects
- System SHALL handle API rate limits (5,000 calls/day for Enterprise edition)

9.8.2 FR-8.2: Billing Platform Integration

- System SHALL ingest data from Stripe/Chargebee: Subscription details, Payment history, Plan changes, Failed charges
- System SHALL process webhook events in real-time

9.8.3 FR-8.3: Analytics Platform Integration

- System SHALL collect product usage data from: Google Analytics (page views, session duration), Mixpanel (event tracking, funnel analysis), Custom analytics APIs

9.9 FR-9: Notification System

9.9.1 FR-9.1: Alert Delivery

- System SHALL support notification channels: Slack (direct message, channel post), Email (digest or real-time), In-app notification center, Microsoft Teams
- Users SHALL configure notification preferences per alert type

9.9.2 FR-9.2: Alert Types

- System SHALL generate alerts for:
 - High-risk account detection
 - Upsell opportunity identification
 - Contract expiration (30/60/90-day reminders)
 - Failed payment events
 - Significant usage decline

9.10 FR-10: Security & Compliance

9.10.1 FR-10.1: Authentication & Authorization

- System SHALL support SSO via OAuth 2.0 (Okta, Azure AD)
- System SHALL enforce role-based access control (RBAC): Admin, Sales Manager, CSM, AE, Read-Only
- System SHALL log all user actions for audit trail

9.10.2 FR-10.2: Data Privacy

- System SHALL encrypt data at rest (AES-256) and in transit (TLS 1.3)
- System SHALL support GDPR/CCPA compliance (data anonymization, deletion)
- System SHALL mask sensitive PII in non-production environments

9.11 FR-11: Autonomous Negotiation & Execution

9.11.1 FR-11.1: Autonomous Engagement

- System SHALL Autonomous Agent capable of initiating and maintaining natural language conversations (email/chat) with customers.
- System SHALL support "Human-in-the-Loop" mode where Agent drafts are queued for approval before sending.
- System SHALL switch to "Fully Autonomous" mode for pre-approved low-risk segments (< \$10k ARR).

9.11.2 FR-11.2: Negotiation Logic

- System SHALL execute negotiation strategies based on playbooks (e.g., offer 5% discount if customer mentions "competitor pricing").
- System SHALL enforce strict guardrails: maximum discount limits, contract term constraints, and required approvals for exceptions.
- System SHALL handle common objections (price, timing, features) using a retrieval-augmented generation (RAG) knowledge base.

9.11.3 FR-11.3: Contract Execution

- System SHALL automatically generate quote documents (PDF) with negotiated terms.
- System SHALL integrate with e-signature platforms (DocuSign/Adobe Sign) to send contracts for signature.
- System SHALL provision licenses/features immediately upon contract signature or payment confirmation.

10 Flow Diagram / Architecture Diagram

10.1 System Architecture Overview

The Renewal & Upsell Advisor follows a microservices architecture with three primary layers:

1. **Integration Layer:** Handles bidirectional data synchronization with external systems (Salesforce, Stripe, Analytics platforms)
2. **Processing Layer:** Executes data transformation, feature engineering, ML inference, and business logic
3. **Presentation Layer:** Provides user interfaces (dashboard) and notification delivery

10.2 Data Flow

Ingestion Phase:

1. External systems (Salesforce, Stripe, Analytics) push data via webhooks or API polling
2. Integration Service validates and queues incoming data
3. ETL Pipeline processes raw data and stores in Data Warehouse

Processing Phase:

1. Feature Engineering Service extracts signals from historical data
2. ML Inference Service calculates risk scores and upsell propensity
3. Playbook Engine matches accounts to recommended plays
4. Results stored in PostgreSQL for real-time access

Action Phase:

1. Notification Service triggers alerts (Slack, Email) based on threshold rules
2. Workflow Automation creates tasks/opportunities in Salesforce
3. Dashboard displays insights and recommendations to users

10.3 Architecture Components

Microservices:

- Integration Service (Node.js / Python FastAPI)
- ETL Pipeline Service (Apache Airflow)
- Feature Engineering Service (Python)

- ML Inference Service (Python + MLflow)
- Playbook Recommendation Engine (Python)
- Notification Service (Python + Celery)
- Dashboard API (FastAPI)
- Frontend Application (React)

Data Stores:

- PostgreSQL: Contracts, accounts, opportunities, tasks
- MongoDB: Event logs, unstructured customer interactions
- Redis: Session cache, API response cache, task queue
- Snowflake/BigQuery: Historical analytics, reporting
- S3: Model artifacts, exported reports

Supporting Infrastructure:

- API Gateway (Kong / AWS API Gateway): Rate limiting, authentication
- Message Queue (RabbitMQ / AWS SQS): Async task processing
- Load Balancer (AWS ALB): Traffic distribution
- Monitoring Stack (Prometheus + Grafana): Metrics, alerts

10.4 Detailed Flow Descriptions

10.4.1 Flow 1: Churn Risk Detection

1. Daily batch job retrieves contracts expiring in 90 days from Salesforce
2. System fetches associated usage data (Google Analytics), payment history (Stripe), support tickets (Zendesk)
3. Feature Engineering Service calculates: 30-day login frequency decline, feature adoption rate, NPS score, payment punctuality score
4. ML Model predicts churn probability for each account
5. Accounts with risk score >70 trigger High Risk alert
6. Playbook Engine recommends intervention playbook
7. Notification sent to assigned CSM via Slack with action items
8. System creates task in Salesforce: "Schedule Executive Business Review"

10.4.2 Flow 2: Upsell Opportunity Identification

1. Real-time webhook from product analytics detects: Account XYZ reaches 85% user license utilization
2. System retrieves account profile: current plan tier, contract value, renewal date
3. Upsell Propensity Model calculates expansion likelihood (78% probability)
4. System estimates expansion value: +25 seats = \$75K additional ARR
5. Opportunity automatically created in Salesforce with "Qualified" stage
6. Personalized email template generated: "Your team is growing fast. Unlock 25 more seats with preferred pricing."
7. AE receives in-app notification with recommended talk track
8. System schedules 3-touch follow-up sequence (Email Day 1, Call Day 3, Email Day 7)

10.4.3 Flow 3: Renewal Forecast Generation

1. Monthly batch job aggregates all renewals for next quarter
2. System calculates weighted renewal forecast: $(\text{Low Risk ARR} \times 95\%) + (\text{Medium Risk} \times 75\%) + (\text{High Risk} \times 40\%)$
3. Expansion pipeline added to base renewal forecast
4. Historical conversion rates applied to upsell opportunities
5. Dashboard displays: Total forecasted ARR, Confidence intervals (P10, P50, P90), Risk-adjusted retention rate
6. Executive report exported to PDF and emailed to VP Sales

Note: Detailed architecture diagram is provided separately as a visual chart. Reference the flow diagram PDF attached with this SRS document.

11 Competitive Analysis

11.1 Market Landscape

The renewal and upsell intelligence market includes specialized customer success platforms, revenue operations tools, and AI-powered sales enablement solutions. Key competitors offer varying degrees of automation, predictive analytics, and CRM integration.

11.2 Primary Competitors

11.2.1 Gainsight Customer Success

Product URL: <https://www.gainsight.com/>

Strengths:

- Industry-leading customer health scoring with ML-powered predictions
- Native Salesforce integration with deep data synchronization
- Comprehensive playbook automation and journey orchestration
- Strong analytics and executive reporting dashboards
- Extensive third-party integrations (200+ apps)

Weaknesses:

- Complex setup requiring significant admin resources
- Higher cost (enterprise pricing starts at \$50K+/year)
- Steeper learning curve for end users
- Limited flexibility for mid-market companies

Technical Approach: Gainsight uses Bayesian recursive models for churn prediction, combining usage telemetry, support data, and sentiment analysis. Their "Rules Engine" triggers automated workflows based on multi-dimensional health scores. Data aggregation occurs in their proprietary data warehouse with scheduled ETL jobs from source systems.

11.2.2 ChurnZero

Product URL: <https://churnzero.com/>

Strengths:

- Real-time customer engagement tracking and alerts
- Easy-to-use interface with faster time-to-value
- Strong renewal forecasting with Forecast Hub
- AI agents (Vibes, Harbinger, Beacon) for risk and opportunity detection
- Competitive pricing for mid-market segment

Weaknesses:

- Less robust analytics compared to Gainsight
- Limited customization for complex enterprise workflows
- Fewer native integrations (relies on Zapier for some connections)
- Reporting can feel rigid per user feedback

Technical Approach: ChurnZero emphasizes real-time event streaming from product analytics. Their AI agents use sentiment shift detection (NLP on emails/chat) and behavioral pattern matching. Renewal forecasting combines health scores with historical win/loss rates. They employ a segment-based approach for playbook recommendations.

11.2.3 Yagna iQ Channel Ecosystem Platform

Product URL: <https://www.yagnaiq.com/>

Strengths:

- **High ROI & Efficiency:** Proven to save 125+ hours per 100 contracts and increase on-time renewal rates by 20–25%.
- **Zero-Touch Automation:** "Set and forget" workflows for low-value, high-volume renewals, triggering proactive quotes 90/60/30 days in advance.
- **Unified Channel View:** Solves the "1000 Portal Problem" by aggregating multi-vendor data into a single dashboard for distributors and resellers.
- **Security & Integration:** ISO 27001 certified with native connectors for Salesforce, Microsoft Dynamics, and major ERPs.

Weaknesses:

- Highly specialized for *indirect* channel sales (Vendors → Distributors → Resellers), which may be overkill for direct-only SaaS businesses.
- UI/UX is function-first (industrial strength) rather than "consumer-grade" aesthetic compared to modern PLG tools.

Technical Approach: Yagna iQ operates as a cloud-native "Channel Ecosystem" platform. It ingests "Golden Record" data from vendor ERPs/CRMs and normalizes it for downstream partners. Its core engine uses AI/ML to identify "Smart Leads" for cross-selling (XSUS Cloud) and automates the entire quote-to-cash cycle for renewals (Renewal Cloud) using event-driven triggers.

11.2.4 Totango

Product URL: <https://www.totango.com/>

Strengths:

- Product adoption-focused with deep usage analytics
- Flexible pricing and modular features (scale as needed)
- Strong customer segmentation capabilities
- Pre-built success plans and templates
- Good balance of simplicity and functionality

Weaknesses:

- Churn prediction less sophisticated than Gainsight
- Limited upsell recommendation intelligence
- Fewer automated playbook options
- Smaller user community and ecosystem

Technical Approach: Totango's core philosophy centers on product usage as the primary health indicator. They auto-generate health scores based on feature adoption patterns without requiring manual configuration. Segments are dynamically updated based on lifecycle stage and account attributes. Renewal risk is inferred from usage decline trends.

11.2.5 Catalyst (formerly ProsperWorks)

Product URL: <https://www.catalyst.io/>

Strengths:

- Unified customer data platform (CDP) approach
- Strong revenue forecasting and pipeline management
- AI-driven insights with "moments" (trigger events)
- Modern UI with intuitive UX

Weaknesses:

- Relatively new player (smaller customer base)
- Fewer proven ML models compared to Gainsight
- Limited third-party integration ecosystem
- Still building out playbook automation

Technical Approach: Catalyst aggregates customer data into a unified profile (CDP model). Their "moments" feature detects trigger events (e.g., usage spike, support escalation) using rule-based and ML hybrid models. Revenue forecasting uses cohort analysis and time-series prediction.

11.3 Innovation & Differentiation Strategy

11.3.1 Core Innovation: The "Agentic" Shift

The primary innovation of our Renewal & Upsell Advisor is the fundamental shift from **Passive Analytics** (Dashboards) to **Active Agency** (Co-Pilots).

1. **From "Insights" to "Actions":** Legacy platforms (Gainsight, Yagna iQ) identify a risk and populate a dashboard, requiring a human Key Account Manager (KAM) to log in, interpret the data, and decide on an action. **Our Innovation:** The Agent identifies the risk and *immediately drafts the remedial action* (e.g., a personalized email, a meeting agenda, or a discount offer) for the KAM to approve. The KAM manages *decisions*, not data.
2. **Generative vs. Static Playbooks:** Competitors use rigid "If-Then" rule engines (e.g., "If health score < 50, send Email Template B"). **Our Innovation:** We utilize Large Language Models (LLMs) to generate *context-aware* playbooks on the fly. The Agent reads the last 10 emails, analyzes the support ticket tone, and generates a truly personalized message that acknowledges specific client pain points, rather than using a generic template.

3. **Unified "Growth Loop" Architecture:** Most platforms treat "Renewals" (Defense) and "Upsells" (Offense) as separate pipelines. **Our Innovation:** We model them as a single **Revenue Optimization Problem**. Every interaction is scored for both Retention Probability and Expansion Propensity simultaneously. The Agent will suppress an upsell offer if the renewal risk is high, or leverage a successful renewal to immediately pitch an upsell, creating a continuous growth loop.

11.3.2 Our Competitive Advantages

1. **Zero-Login Experience:** The Agent lives where the KAM works (Slack, Teams, Email). There is no "platform" to learn or log into.
2. **Hyper-Personalization at Scale:** Using Generative AI to craft unique account plans for thousands of long-tail customers that would otherwise receive generic automation.
3. **Faster Time-to-Value:** 7-day deployment via API overlay, compared to 6–12 week implementation cycles for heavy platforms like Yagna or Gainsight.

11.4 Feature Comparison Matrix

Feature	Our Agent	Gainsight	ChurnZero	Totango	Catalyst	Yagna iQ
Churn Prediction	Action-Oriented	Descriptive	Descriptive	Basic	Descriptive	Analytical
Upsell Propensity	Generative Pitch	Partial	Yes	Limited	Yes	Smart Leads
Playbook Engine	GenAI (Dynamic)	Rigid Rules	Rigid Rules	Rigid Rules	Rigid Rules	Rigid Workflow
Real-Time Alerts	Drafts Response	Notification	Notification	No	Notification	Quote Trigger
User Interface	Chat/Slack (Headless)	Dashboard	Dashboard	Dashboard	Dashboard	Portal
Salesforce Sync	bidirectional Agent	Native Sync	Sync	Sync	Native Sync	Native Sync
Deployment	7 Days (Overlay)	8–12 weeks	4–6 weeks	2–4 weeks	3–5 weeks	4–8 weeks
Pricing	Usage/Outcome	Seat-Based	Seat-Based	Seat-Based	Seat-Based	Volume-Based
Core AI Tech	LLM + RAG	Predictive ML	Predictive ML	Basic ML	Predictive ML	Predictive ML

11.5 Market Presence & Feedback

Platform	Est. Customers	Revenue (Latest Est.)	Review Highlights
Our Agent	Emerging; no public count (Sutherland-linked for contact centers)	Not disclosed	Limited reviews; praised for AI agent success in ops

Gainsight	Thousands (nearly 200 public enterprises; segmented by revenue tiers)	\$272M ARR	Strong for CS scalability (NPS 85); top-rated in benchmarks
ChurnZero	Hundreds (mid-market focus, e.g., Imagine Learning)	Not disclosed (\$50-100M inferred from market share)	Good real-time alerts; 4.5/5 avg. for usability
Totango	5,000+ businesses (incl. Zoom, Google)	Not disclosed	Solid AI forecasting; mixed on pricing (4/5 avg.)
Catalyst	No specific count (CS platform focus)	Not disclosed	Basic features; limited feedback available
Yagna iQ	30 vendors, 5K resellers (channel-focused)	Not disclosed	Positive for CPQ/sales streamlining; niche appeal

11.6 References

1. Gainsight Customer Success Platform: <https://www.gainsight.com/customer-success/>
2. ChurnZero Renewal Management: <https://churnzero.com/renewal-expansion-solutions/>
3. Totango vs Gainsight Comparison: <https://www.accoil.com/blog/totango-gainsight>
4. Customer Success Renewal Forecasting: <https://churnzero.com/blog/customer-success-renewal-forecasting/>
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6. ML-Based Renewal Prediction: <https://www.sirion.ai/library/contract-insights/predicting-renewal-probability-machine-learning/>
7. Stripe + Salesforce Integration Guide: <https://candyboxcrm.com/blog/how-to-set-up-the-stripe-salesforce-integration/>
8. ARR Forecasting Methods: <https://www.drivetrain.ai/post/arr-forecasting-corkscrew-r/>
9. Churn Prediction Tools 2025: <https://www.zendesk.com/in/service/customer-experience/churn-prediction-software/>
10. Sentiment Analysis for Churn: <https://insight7.io/how-to-use-sentiment-analysis-to-pr/>

12 Assumptions & Constraints

12.1 Assumptions

1. Client has active Salesforce instance (Enterprise or Unlimited edition).
2. Historical data available for minimum 2 years (contracts, renewals, usage).
3. API access credentials provided for all integrated systems.
4. Dedicated technical point-of-contact available during development.
5. Cloud infrastructure budget allocated (AWS/GCP).

12.2 Constraints

1. 7-day development timeline (no scope creep).
2. API rate limits from external systems (Salesforce: 5,000 calls/day).
3. Data privacy regulations (GDPR, CCPA compliance required).
4. Limited budget for third-party ML services (prefer open-source).
5. Single deployment environment (production only, no separate staging initially).

13 Success Metrics

13.1 Technical KPIs

- **ML Model Accuracy:** >85% precision and recall on churn prediction
- **System Uptime:** 99.5% availability
- **API Response Time:** <200ms (P95)
- **Data Sync Latency:** <5 minutes for real-time updates
- **Alert Delivery Time:** <60 seconds from trigger event

13.2 Business KPIs

- **Churn Reduction:** 20-30% decrease in revenue churn
- **Expansion Revenue:** 15-25% increase in upsell conversions
- **Forecast Accuracy:** >90% accuracy on quarterly renewal predictions
- **Team Productivity:** 35% reduction in manual renewal tracking time
- **Early Intervention Rate:** 80% of high-risk accounts engaged within 5 days of alert

14 Risk Management

14.1 Technical Risks

- **Risk:** API rate limit exceeded during peak usage
 - **Mitigation:** Implement request queuing, caching, and batch processing
- **Risk:** ML model drift (accuracy degrades over time)
 - **Mitigation:** Automated weekly model performance monitoring, quarterly re-training
- **Risk:** Integration failure with external systems

- **Mitigation:** Implement retry logic, fallback mechanisms, comprehensive error logging

14.2 Business Risks

- **Risk:** User adoption challenges (sales team ignores recommendations)
 - **Mitigation:** Comprehensive training, in-app guidance, demonstrate early wins
- **Risk:** Data quality issues (incomplete or inaccurate source data)
 - **Mitigation:** Data validation rules, quality scoring, alert admins to data gaps

15 Appendices

15.1 Appendix A: Glossary

- **ARR:** Annual Recurring Revenue
- **MRR:** Monthly Recurring Revenue
- **Churn:** Loss of customers or revenue
- **NPS:** Net Promoter Score
- **CSM:** Customer Success Manager
- **AE:** Account Executive
- **CRM:** Customer Relationship Management
- **ETL:** Extract, Transform, Load
- **ML:** Machine Learning
- **NLP:** Natural Language Processing

15.2 Appendix B: API Endpoints (Sample)

```
POST /api/v1/predict/churn
GET /api/v1/renewals?risk_level=high
POST /api/v1/playbooks/recommend
GET /api/v1/opportunities/upsell
POST /api/v1/alerts/configure
```

15.3 Appendix C: Contact Information

- **Project Manager:** [Name], [Email]
- **Tech Lead:** [Name], [Email]
- **Product Owner:** [Name], [Email]

16 Document Approval

Prepared By: Date:	[Name], [Title]
Reviewed By: Date:	[Name], [Title]
Approved By: Date:	[Name], [Title]